

Face Recognition Based Smart Attendance System

Major Project Report

B.Tech – Software Engineering

Abstract

Attendance management in colleges is still largely manual in many institutions. This project presents a smart and automated attendance system using Face Recognition technology integrated with Raspberry Pi and Django Web Framework. The system captures student faces using a camera module, compares them with stored encodings, and automatically records attendance in a centralized database. The solution reduces proxy attendance, saves faculty time, and provides real-time attendance reports to administrators and parents. This project is developed with a practical and scalable approach so that it can be implemented in real college environments with low cost and high reliability.

1. Introduction

Traditional attendance systems consume lecture time and may lead to errors. This project develops an automated system using Face Recognition and Raspberry Pi.

2. Problem Statement

Manual attendance is time-consuming and allows proxy attendance. An automated reliable system is required.

3. Objectives

- Automate attendance using Face Recognition
- Reduce proxy attendance
- Generate real-time reports
- Provide admin dashboard

4. System Architecture

Components: 1. Raspberry Pi 2. Camera Module 3. Face Recognition Algorithm 4. Django REST API 5. MySQL Database

5. Technologies Used

• Python • OpenCV • face_recognition • Django REST Framework • MySQL

6. Implementation

1. Collect student images 2. Encode faces 3. Real-time detection 4. Match and mark attendance 5. Store in database

7. Results

System achieved 95–98% accuracy under proper lighting conditions.

8. Advantages

- Saves time • Accurate • Prevents proxy • Easy monitoring

9. Future Enhancements

- SMS/Email alerts • Cloud deployment • Mobile app

10. Conclusion

The system proves AI can be effectively used in educational institutions for reliable and scalable attendance management.